॥ श्री गणेशाय नमः ॥

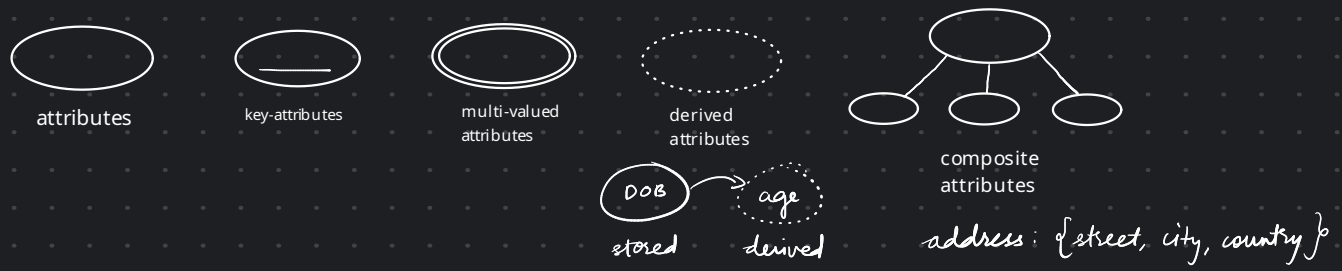
# ER Model

## Entity-relationship model

- a high-level diagrammatic representation of DB design

Steps to design a db:
- requirements
- conceptual/logical db design → ER
- apply normalization
- physical db design
- application/security

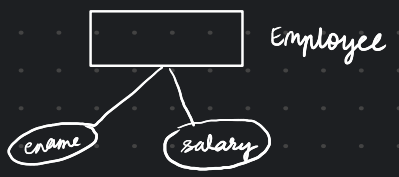
## Attributes




## Entity set

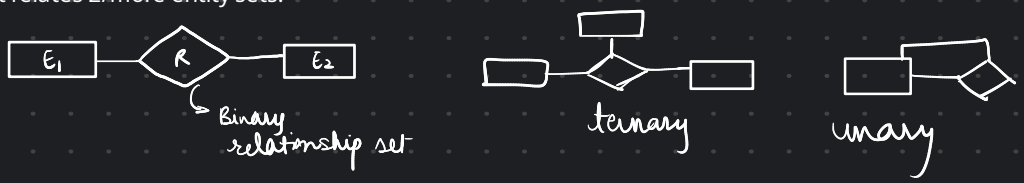
→ Think of it as a relation/ table

An entity set is a set of similar entities (tuples).




## Relationship set

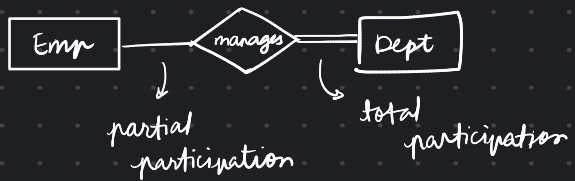
It relates 2/more entity sets.

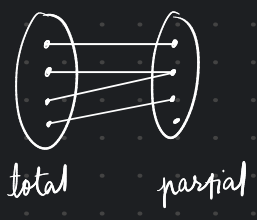



## Participation

If every entity of an entity set participates in a relationship set, it is said to be in total participation.
Otherwise, it is said to be in partial participation.

E.g. Emp and Dept are entity sets. MANAGES is a relationship set such that there should be a manager for every Dept.



## Cardinality



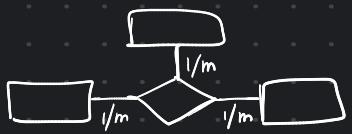

- One (atmost 1) ⟶ 0/1
- Many (0 or more) ——— 0…* ——— m

1 to many { 1 Emp manages 0/more Depts
many to one { 1 Dept has atmost 1 Emp as manager } will be exactly 1 due to total participation
Total { All Depts have an Emp as manager

## Binary relationship sets




## Ternary relationship sets




# of such sets = 2 × 2 × 2 = 8

## One-to-many mapping






An element of the first set can be mapped to many elements of the second, but one element of the second set can only be mapped to atmost 1 element of the first set.

1. All the keys of the entity sets should be present in the entity set.

$E_1$

| A | B |
|---|---|
| $a_1$ | – |
| $a_2$ | – |
| $a_3$ | – |

$A \to B$

R

| A | C |
|---|---|
| $a_1$ | $c_1$ |
| $a_1$ | $c_2$ |
| $a_2$ | $c_3$ |

↳ A repeats, C doesn't.

$\therefore C \to A$

$E_2$

| C | D |
|---|---|
| $c_1$ | – |
| $c_2$ | – |
| $c_3$ | – |
| $c_4$ | – |

$C \to D$

For 1 to many relationships,
key of the RHS will become the candidate key for the relationship set.
↳ "many" side

3 tables are sufficient. Are they necessary?

The attributes are unique. We could do this in 2 tables.

$E_1$

| A | B |
|---|---|
| $a_1$ | |
| $a_2$ | |
| $a_3$ | |
| $a_4$ | |

$A \to B$

$RE_2$

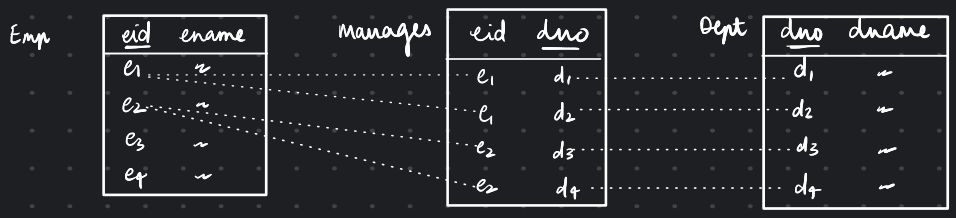
| C | D | A |
|---|---|---|
| $c_1$ | – | $a_1$ |
| $c_2$ | – | $a_1$ |
| $c_3$ | – | $a_2$ |
| $c_4$ | – | NULL |

$C \to A$, $C \to D$ } $C \to AD$

For a one-to-many relationship, regardless of the participation
2 tables are sufficient.
The relationship merges on the "many" side.

E.g. "Emp" and "Dept" are 2 entities and "manages" is a 1:m relationship set between them. Each Emp can manage many Depts but each Dept must be managed by exactly 1 Emp.




can be merged to

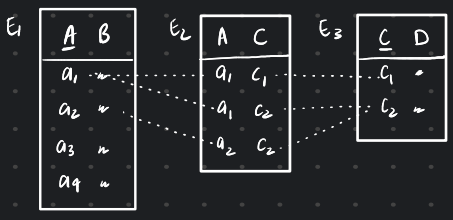



the relationship set is merged into the "many" side

For 1:m mapping, the relationship set combines into the 'many' set.
If there's total participation, we won't get any NULL values.
If there's partial participation, there may be NULLs.

## Many-to-many mapping



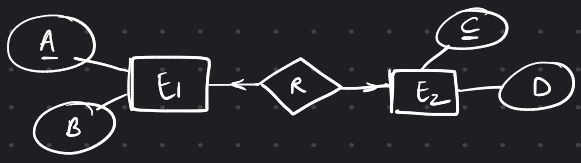

If R is merged into E1, A won't remain the primary key as for 1 A value a1, there are corresponding 2 C values, c1 and c2.
Similarly, R cannot be merged into E2, as for 1 C value c2, there are corresponding 2 A values, a1 and a2.

For a many-to-many relationship mapping, we need 3 tables at least (E1, R, and E2).
We need 2 foreign keys to maintain the foreign key constraints. → else the candidate key is violated

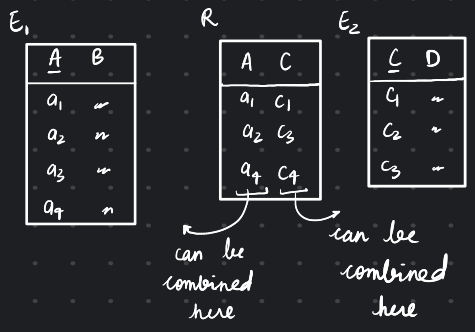
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## One-to-one mapping

1:1 mapping between E1 and E2 with partial participation at both ends:




1 instance of E1 is related to atmost 1 instance of E2. 1 instance of E2 is related to atmost 1 instance of E1.

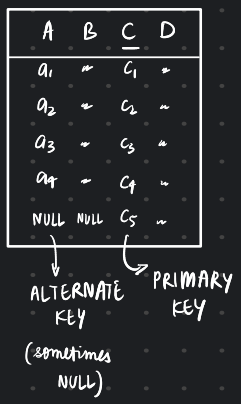



In the relationship set, the 2-tuple is unique.
We could merge the relationship table into either of the 2 entity tables.

1:1 mapping between E1 and E2 with at least 1 end in total participation:




1 of the sides is in total participation in the 1:1 mapping. We could combine all of them into a single table.



1:1 mapping between E1 and E2 with both ends in total participation:

A subset of the above case. 1 table is sufficient to represent the mapping.




| A | B | C | D |
|---|---|---|---|
| $a_1$ | " | $c_1$ | " |
| $a_2$ | " | $c_2$ | " |
| $a_3$ | " | $c_3$ | " |
| $a_4$ | " | $c_4$ | " |

either of these can be the CK.

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## Self-referential relationship sets

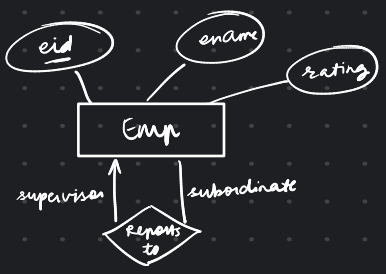
Entities of entity set (E) related to some of the entities of the same entity set (E) are in a self-referential or unary or recursive sets.




### 1:many self-referential relationships

E.g. the "Emp" entity set and "reports to" relationship set are related (between supervisors and sub-ordinates). Each superviser can supervise many sub-ordinates and each sub-ordinate reports to at most 1 supervisor.

supervisors and subordinates, both are instances of the "Emp" entity set.




Emp

| eid | ename | rating |
|---|---|---|
| $e_1$ | | |
| $e_2$ | | |
| $e_3$ | | |
| $e_4$ | | |

Reports_to

| supervisor | subordinate |
|---|---|
| $e_1$ | $e_2$ |
| $e_1$ | $e_3$ |
| $e_2$ | $e_4$ |

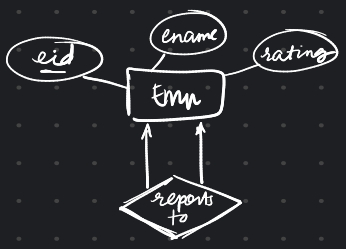
not repeating

We can combine these into 1 table.

| eid | ename | rating | supervisor |
|---|---|---|---|
| $e_1$ | | | " |
| $e_2$ | | | $e_1$ |
| $e_3$ | | | $e_1$ |
| $e_4$ | | | $e_2$ |

### 1:1 self-referential relationships

In the above example, consider that each superviser can supervise only 1 subordinate.



Emp

| eid | ename | rating |
|---|---|---|
| $e_1$ | | |
| $e_2$ | | |
| $e_3$ | | |
| $e_4$ | | |

Reports_to

| sup | sub |
|---|---|
| $e_1$ | $e_2$ |
| $e_2$ | $e_3$ |
| $e_3$ | $e_4$ |

can be combined to :

Emp

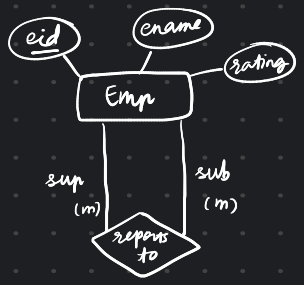
| eid | ename | rating | sup |
|---|---|---|---|
| $e_1$ | | | NULL |
| $e_2$ | | | $e_1$ |
| $e_3$ | | | $e_2$ |
| $e_4$ | | | $e_3$ |

OR

| eid | ename | rating | sub |
|---|---|---|---|
| $e_1$ | | | $e_2$ |
| $e_2$ | | | $e_3$ |
| $e_3$ | | | $e_4$ |
| $e_4$ | | | NULL |

## many:many self-referential relationships

In the above example, consider that each superviser can supervise many subordinates and each subordinate can report to many supervisors.




Emp

| eid | ename | rating |
|---|---|---|
| $e_1$ | | |
| $e_2$ | | |
| $e_3$ | | |
| $e_4$ | | |

Reports_to

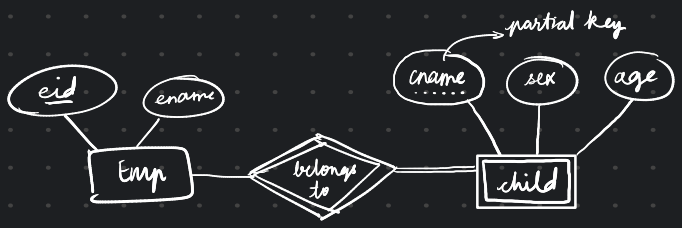
| supervisor | subordinate |
|---|---|
| $e_1$ | $e_2$ |
| $e_1$ | $e_3$ |
| $e_4$ | $e_3$ |

We can't really combine these into a single table. We'll need 2 tables and 2 foreign keys as well.

---

## Weak entity sets 

These are entity sets with no keys, i.e. attributes of weak entity sets are not sufficient to differentiate records uniquely.
Weak entities are depending entities which depend on other strong entities sets (called identifiers or owners).




Emp

| eid | ename |
|---|---|
| $e_1$ | |
| $e_2$ | |
| $e_3$ | |
| $e_4$ | |

| eid | cname |
|---|---|
| $e_1$ | A |
| $e_2$ | A |
| $e_2$ | B |
| $e_3$ | B |

child

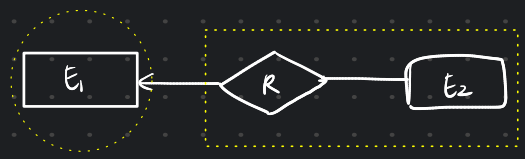
| cname | sex | age |
|---|---|---|
| A | | |
| A | | |
| B | | |
| B | | |

the same thing
(can be merged)

Both together are the candidate key

| cname | eid | sex | age |
|---|---|---|---|
| A | $e_1$ | | |
| A | $e_2$ | | |
| B | $e_2$ | | |
| B | $e_3$ | | |

# Summary

1. 

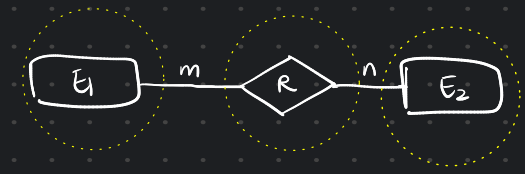

min RDMBS tables = 2
min FKs = 1

1:m
partial-partial

2. 

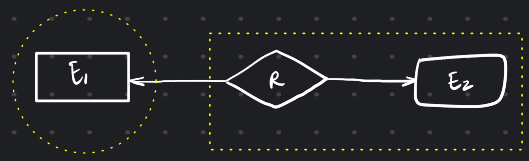

min RDMBS tables = 2
min FKs = 1

1:m
partial-total

3. 


min RDMBS tables = 3
min FKs = 2

m:n
partial-partial

4. 

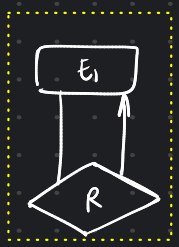

min RDMBS tables = 3
min FKs = 2

1:1
partial-partial

5. 


min RDMBS tables = 1
min FKs = 0

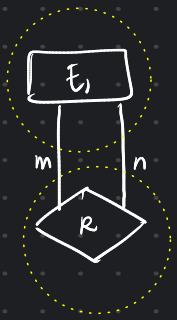
1:1
at least 1 total
(or both total)

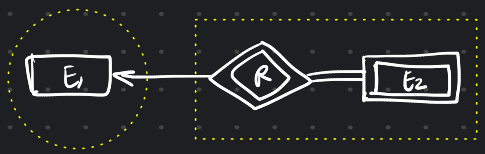
6. 


min RDMBS tables = 1
min FKs = 0

1:1
at least 1 total
(or both total)

7. 


8. 


9. 


min RDMBS tables = 2
min FKs = 1

PYQs

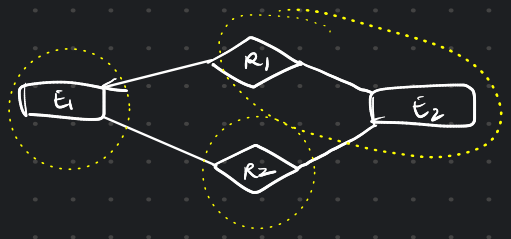
**3.6.1 Er Diagram: GATE CSE 2005 | Question: 75** top ★ https://gateoverflow.in/1398

Let $E_1$ and $E_2$ be two entities in an $E/R$ diagram with simple-valued attributes. $R_1$ and $R_2$ are two relationships between $E_1$ and $E_2$, where $R_1$ is one-to-many and $R_2$ is many-to-many. $R_1$ and $R_2$ do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?

A. 2
B. 3
C. 4
D. 5

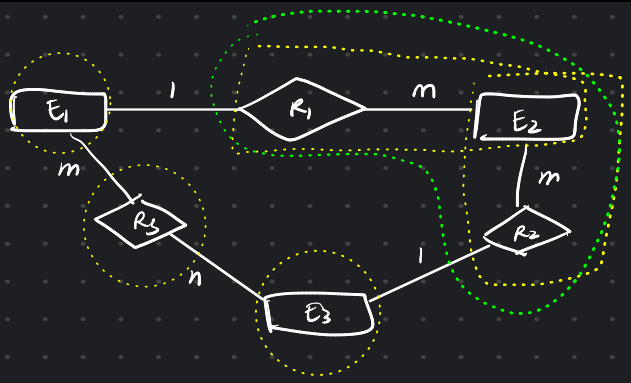
tests.gatecse.in goclasses.in tests.gatecse.in

gate2005-cse databases er-diagram normal




3

Q) $E_1$ $E_2$ $E_3$ entity sets. $R_1$ $R_2$ $R_3$ relationship sets. $R_1$ is related b/w $E_1$ and $E_2$ with 1:m. $R_2$ is related b/w $E_2$ and $E_3$ with m:1. $R_3$ is related b/w $E_1$ & $E_3$ with m:N.




4